TherapEase - A Virtual Therapist Assistant Digital Twin

Project Team

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Chapter 1

Introduction

This project aims to

- develop a Therapist Assistant (a 3D Digital Twin)
- assist therapists in monitoring the progress of autistic individuals
- improving engagement of autistic individuals
- accurately assessing emotional responses before therapy sessions

Most of the current therapy methods for individuals with Autism Spectrum Disorder (ASD) rely on manual assessments by therapists, which are time-consuming and not always reliable, making it difficult to monitor small short-term changes in behavior or emotions. Consistent patient engagement through therapy is also needed. We will automate this by creating a Digital Twin of a Therapist Assistant, assisting with real-time emotion detection, personalized therapy, and automatic progress reports. This document specifies the requirements for developing a web-based application that serves as a therapist assistant for individuals with ASD. The system features a 3D digital twin to engage users, emotion detection, and Automated Diagnostic Decision Support (ADDS) scores based on responses to interactive questions. Therapists will have a dashboard displaying session data, including emotions, responses, and ADDS scores.

1.1 Problem Statement

TherapEase is being developed to support therapists working with autistic individuals by assessing them while keeping them engaged in order to improving therapy outcomes. Therapists currently rely on manual methods to assess the progress of individuals with Autism Spectrum Disorder (ASD). These methods make it difficult to track progress and accurately understand emotional states between therapy sessions. This can create gaps in understanding the patient's development, which may reduce the overall effectiveness of the therapy. TherapEase addresses these challenges by providing an interactive and personalized environment where patients can feel comfortable expressing their emotions. By incorporating AI techniques, the system can detect emotional responses, allowing therapists to adjust their approach as needed. Conclusively, the goal is to create a supportive tool that enhances therapists experience for treating autistic individuals.

1.2 Scope

The scope of TherapEase is to develop a 3D digital twin of a therapist assistant and digitize the progress tracking of autistic individuals. The application will have two login systems: one for therapists and one for patients. Therapists will access a dashboard to monitor individual progress, including emotion detection results, therapy scores, and patient data, allowing them to adjust therapy plans. The interactive environment for patients will feature a human 3D digital twin that asks questions and detects emotions and behaviors during assessments, passing data to an AI model for further analysis. Scores will be generated based on interactions and emotion detection, contributing to an approximate rating for progress evaluation. These scores will be stored in a database for each patient. The project focuses on the behavioral domain, specifically ABA assessments[1]. Common methods include ESDM, VB-MAPP, ABLLS-R, PEAK, EFL, and AFLS[2]. The age limit for patients is 12+, as diagnostic tools can be less accurate for younger children. Detected emotions during assessment include happiness, sadness, anger, surprise, fear, and disgust. Videos will be recorded after each question, with separate models for eye blinking and emotion detection.

Exclusions:

- VR-based therapy environments.
- No focus on hardware development beyond webcams and computers.
- No integration with existing EHR systems unless specified for future developments.

1.3 Modules

1.3.1 Module 1

In this module, we will clearly define scope and requirements of our project along with the objectives, target users and core functionalities of the system. The Software Requirements

Specification(SRS), including Functional and Non-Functional Requirements along with the Software Development Plan, will be included in Module 1. It also includes gathering essential data for autistic individuals like questions which will be asked from them for assessment. We will be working on prototype of our project, Final Year Project Poster along with the diagrams like class-domain and activity diagrams etc.

Deliverables:

1. Define scope and requirements
2. Design overall Pipeline
3. Data Collection and Preparation

1.3.2 Module 2

This module mainly focuses on creation of digital twin 3D basic model which is virtual assistant of the therapist. This model will interact with the patients in the 3D virtual environment. In addition to that, we will also be training AI models in this module for emotion detection and natural language processing to improve interaction and engagement of a patient during therapy.

Deliverables:

1. Create basic digital twin 3D model
2. Develop initial CV, GenAI and NLP models

1.3.3 Module 3

We will be focusing on implementing the remaining features of our application during this module. We will work on different interactive 3D environments where our digital twin model will be placed. In addition, a score system will be created that analyzes interactions, facial expressions, and other behaviors identified by our previously developed AI models. A database management system (DBMS) will also be designed in this module to manage our project's storage and retrieval features. Finally, we will create a dashboard, making it easy for therapists to keep track of the scores.

Deliverables:

1. 3D environment designs
2. Scoring algorithm

3. Implement DBMS
4. Dashboard Creation

1.3.4 Module 4

The final module consist of deploying the system and continuous testing to make sure everything is fine. Any errors or issues are resolved during this phase. The user interface is made user friendly so it is easy to use, which gives therapists quick access to patient records.

Deliverables:

1. Deployment
2. Testing and Refinement

1.4 User Classes and Characteristics

User class	Description
Therapists	Therapists and clinical psychologists are trained professionals who are specialised in working with autistic individuals. They mostly use traditional therapy methods and assessments in order to provide proper support to their patients treat. They seek tools that facilitate efficient tracking of patient progress so they can tailor their approaches based on collected relevant data.
Patients (Autistic Individuals)	The patients using the TherapEase system will mostly be autistic individuals aged 12 years and above. Most of them may have varying levels of emotional awareness and communication skills, and this will impact their interaction with the therapy platform. Hence, it is important to create an engaging environment, allowing the patient to say whatever he/she feels freely.
Developer/ Administrators	Administrators are responsible for the overall management of the TherapEase platform, mostly managing user access, ensuring data privacy regulations, and maintaining security measures to protect sensitive information. Administrators will have access to the application in order to monitor the performance of the system, identifying areas for improvement. They are also there for providing support to therapists, caregivers, and other users, helping them navigate the platform's features effectively.
Researchers	Researchers studying autism and therapeutic methods will also find value in the TherapEase platform. They typically possess strong technical skills for data analysis, allowing them to manipulate and interpret complex datasets so they can use TherapEase to explore trends, correlations, and outcomes, in order to contribute to the field of autism research.

Table 1.1: User Classes

Chapter 2

Project Requirements

2.1 Use-case

2.1.1 Manages Patient Profiles

Actor: Therapist
Description: The therapist manages patient profiles by adding, viewing, or deleting patient information.
Preconditions: The therapist is logged into the system.
Steps:

1. The therapist selects the "Add patient" or "View Patient Report" or "Delete Patient" option.

2. The system processes the request and updates the patient profiles accordingly.

Postconditions: Patient profiles are updated in the system (added, viewed, or deleted).

2.1.2 Views Progress Reports

Actor: Therapist
Description: The therapist views progress reports for each patient to track therapy outcomes.
Preconditions: The system contains progress reports for the patient.
Steps:

1. The therapist navigates to the "View Patient Reports" section.

2. The therapist enters a patient MRNO.

3. The system displays the patient's progress report based on past therapy sessions.

Postconditions: The therapist can review the patient's progress report.

2.1.3 Modifies Therapy Plans

Actor: Therapist
Description: The therapist modifies therapy plans based on patient progress and therapy outcomes.
Preconditions: The therapist has access to the patient's progress reports.
Steps:

1. Its the therapist job to see the report generates and then proceed with whatever therapy plans he wants to execute in the physical session.

Postconditions: Therapist can set his Therapy plans easily.

2.1.4 Conducts Emotion Detection

Actor: System
Description: The system conducts emotion detection during therapy sessions to assess the patient's emotional state.
Preconditions: The system must have access to real-time video or audio input for emotion analysis.
Steps:

1. The system runs emotion detection algorithms on the patient's live video or Video feed.
2. The system analyzes facial expressions.
3. The system logs the detected emotions.

Postconditions: The emotional data is stored in the system for further analysis.

2.1.5 Stores Therapy Scores

Actor: System
Description: The system stores therapy scores based on the patient's performance during sessions.
Preconditions: The patient must have completed a therapy session.
Steps:

1. After the therapy session, the system calculates the therapy scores.
2. The system stores these scores along with the session data.

Postconditions: The therapy scores are saved in the patient's profile for future reference.

2.1.6 Manages Patient Records

Actor: System
Description: The system manages and maintains patient records, including personal details and therapy history.
Preconditions: Patient data must already be available in the system.
Steps:

1. The system collects and organizes the patient's therapy data.
2. It updates patient records after each therapy session.

Postconditions: The patient's records are up to date with the latest therapy information.

2.1.7 Displays 3D Digital Twin

Actor: System
Description: The system displays a 3D digital twin during therapy sessions to interact with the patient.
Preconditions: The system must have a 3D digital twin model ready to be shown to the patient.
Steps:

1. The system displays the 3D digital twin on the screen during the therapy session.
2. The digital twin engages the patient, asking questions.

Postconditions: The 3D digital twin assists in the therapy session.

2.1.8 Interacts with 3D Digital Twin

Actor: Patient
Description: The patient interacts with the 3D digital twin during therapy sessions.
Preconditions: The system must have displayed the 3D digital twin.
Steps:

1. The 3D digital twin displays questions or activities to the patient.
2. The patient interacts by answering the questions or performing actions.

Postconditions: The system logs the patient's interaction for later analysis.

2.1.9 Answers Questions

Actor: Patient
Description: The patient answers questions posed by the 3D digital twin during the therapy session.
Preconditions: The system has displayed the 3D digital twin and posed questions to the patient.
Steps:

1. The 3D digital twin asks questions.
2. The patient answers the questions verbally or through a provided interface.
3. The system processes and records the patient's answers.

Postconditions: The patient's responses are logged in the system for analysis or report generation.

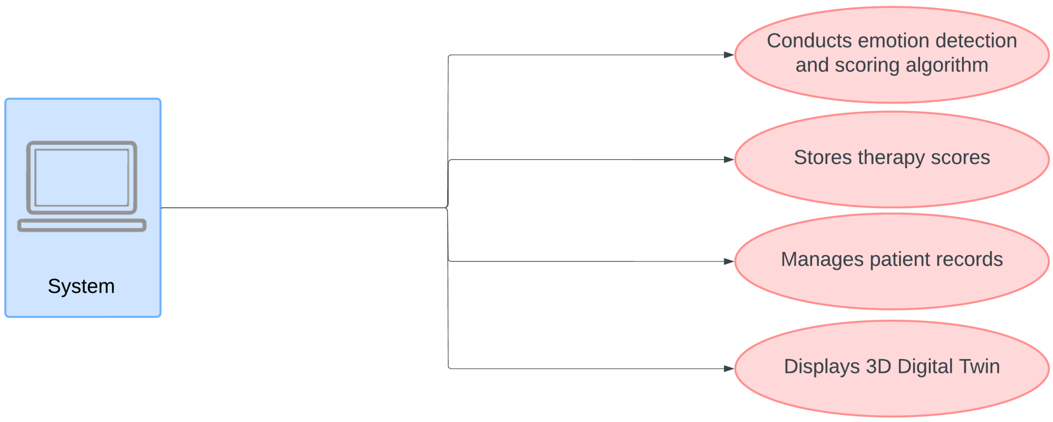

Figure 2.1: Use Case Diagram for System

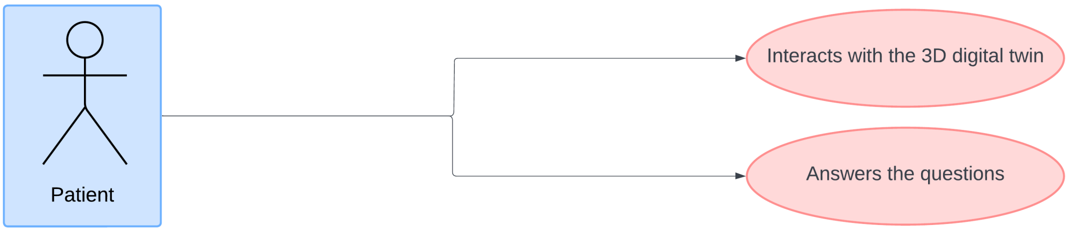

Figure 2.2: Use Case Diagram for Patient

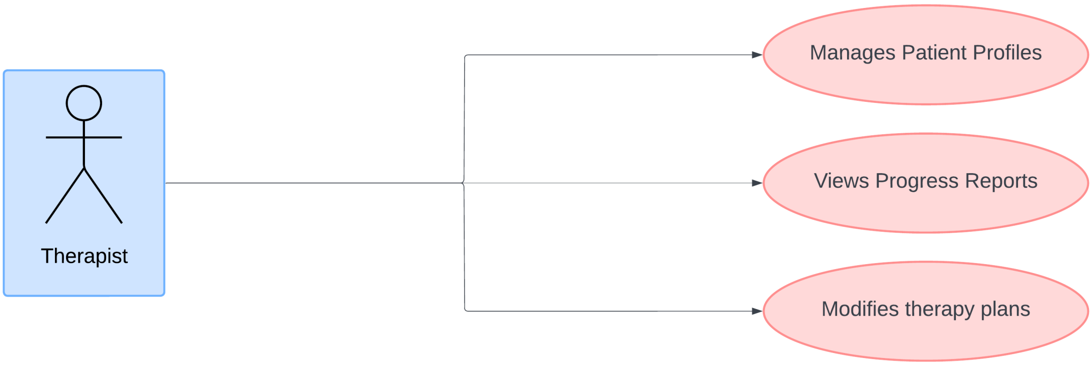

Figure 2.3: Use Case Diagram for Therapist

2.2 Definitions, Acronyms, and Abbreviations

- **ASD**: Autism Spectrum Disorder
- **EFL** : Essential for Living
- **ADDS**: Automated Diagnostic Decision Support
- **ESDM** : Early Start Denver Model
- **PEAK** : Promoting Emergence of Advanced Knowledge
- **AFLS** : Assessment of Functional Living Skills
- **VB-MAPP** : Verbal Behavior MileStones Assessment Placement Program
- **ABLLS-R** : Assessment of Basic Language and Learning Skills-Revised
- **3D Digital Twin**: A virtual human avatar interacting with users in real time

- **Emotion Detection**: The process of detecting facial expressions and emotional states using AI models
- **Dashboard** : A graphical interface that presents data to therapists for reviewing patient progress
- **Therapist Assistant**: The functionality of the app to assist therapists by storing and analyzing user data

2.3 Functional Requirements

2.3.1 User Interaction with 3D Digital Twin

- **Description**: The application will display a 3D digital twin to interact with users during therapy.
- **Inputs**: User responses to prompts (e.g., object selection, color choices).
- **Outputs**: Animated responses from the digital twin and recorded user interactions.
- **Functional Requirements**:
  - The digital twin should engage users with interactive prompts and guide them through therapy sessions.
  - Users can perform tasks like selecting shapes, colors, or answering yes/no questions.

2.3.2 Emotion Detection

- **Description**: Detect the user's emotional state (e.g., happiness, sadness, confusion) during their interactions with the digital twin.
- **Inputs**: video of the user.
- **Outputs**: Detected emotional state (stored in the database).
- **Functional Requirements**:
  - The system should use a facial emotion detection model to capture the user's emotional state.
  - Emotion data should be updated and stored for therapist review.

2.3.3 ADDS Score Calculation

- **Description**: Generate ADDS scores based on user responses during the session.
- **Inputs**: User responses to various tasks (e.g., object identification, color selection).
- **Outputs**: ADDS score stored in the database.
- **Functional Requirements**:
  - The system should calculate ADDS scores using predefined algorithms based on the user's performance.
  - Scores should be generated and stored automatically at the end of each session.

2.3.4 Database Storage

- **Description**: Store user responses, detected emotions, and ADDS scores in a secure database.
- **Inputs**: User interaction data, emotion detection results, ADDS scores.
- **Outputs**: Stored data accessible to therapists.
- **Functional Requirements**:
  - All data must be stored securely in compliance with privacy standards.
  - Therapists should have access to historical data for review and analysis.

2.3.5 Therapist dashboard

- **Description**: Provide therapists with a visual summary of user responses, emotions, and ADDS scores.
- **Inputs**: Stored data from therapy sessions.
- **Outputs**: Graphical dashboard showing session history, ADDS scores, and emotional changes.
- **Functional Requirements**:
  - The dashboard should show trends in the user's emotional state and ADDS scores across multiple sessions.
  - Therapists can filter data based on session dates or specific tasks.

2.4 Non-Functional Requirements

2.4.1 Reliability

The web app must have an uptime of at least 99%, ensuring it is available when users need it.

2.4.2 Usability

The interface should be intuitive and accessible for users with cognitive impairments.

2.4.3 Performance

The system must be capable of processing and storing large amounts of data (e.g., video streams, emotion data).

2.4.4 Security

The system must contain only users (Patients) that have admin (Therapist) permissions to protect user data.

2.5 Domain Model

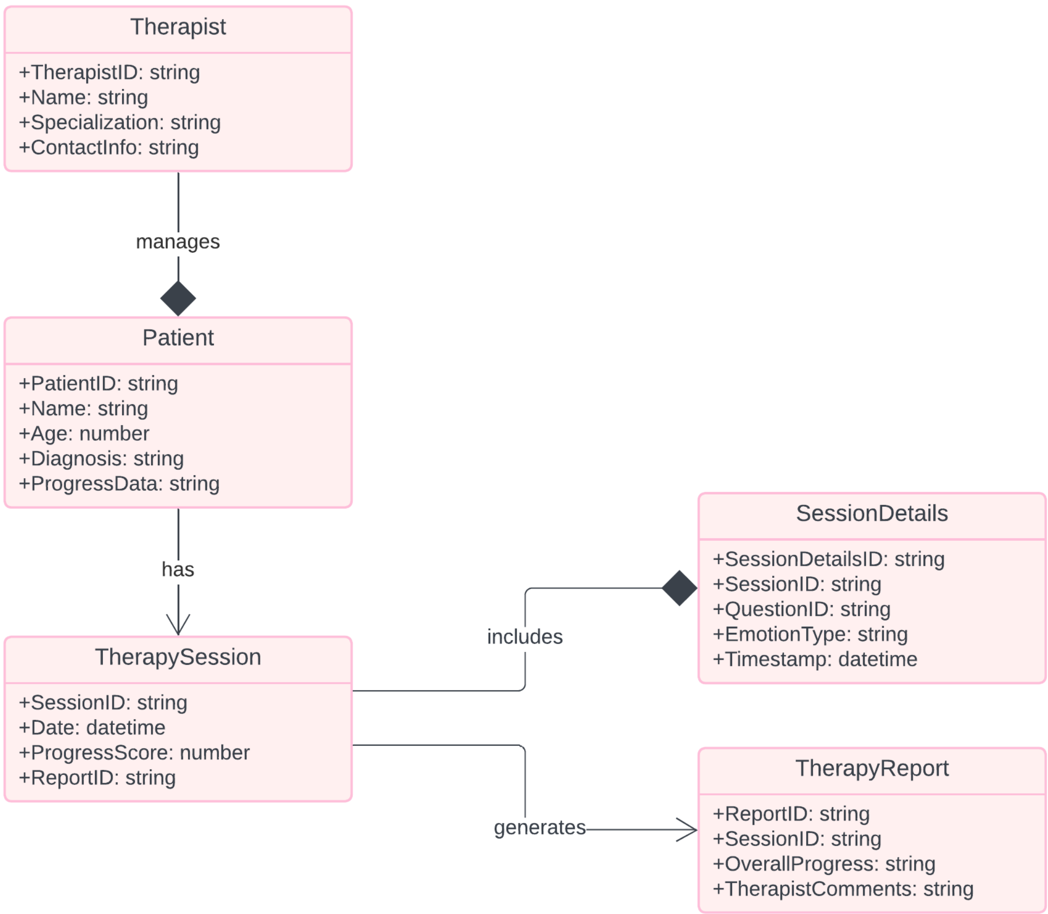

Figure 2.4: Domain Model

Chapter 3

System Overview

The functionality of this system is providing emotion detection, engagement tracking, and diagnostic scoring through an interactive 3D environment. Therapists will be able to access a dashboard in order to monitor patient progress and emotional states, while patients engage with a digital twin to for assessment.

The system uses AI models to detect emotions like happiness, sadness, anger, and surprise, and process those to generate scores which assist therapists. TherapEase aims to enhance therapy effectiveness by automating assessments during the gaps between therapy sessions, offering a more consistent approach to ASD care.

3.1 Architectural Design

- **Frontend (UI)**:
  - Therapist Dashboard
  - Patient Interface (for interactive sessions)
  - 3D Environment (Unity and Blender)
- **Backend**:
  - Core system logic and connection to AI models
  - ADDS scoring model to track diagnostic metrics
  - Session and user data management
- **Database**:
  - Stores patient data, session logs, therapy scores, and emotion analysis results
- **AI Models**:

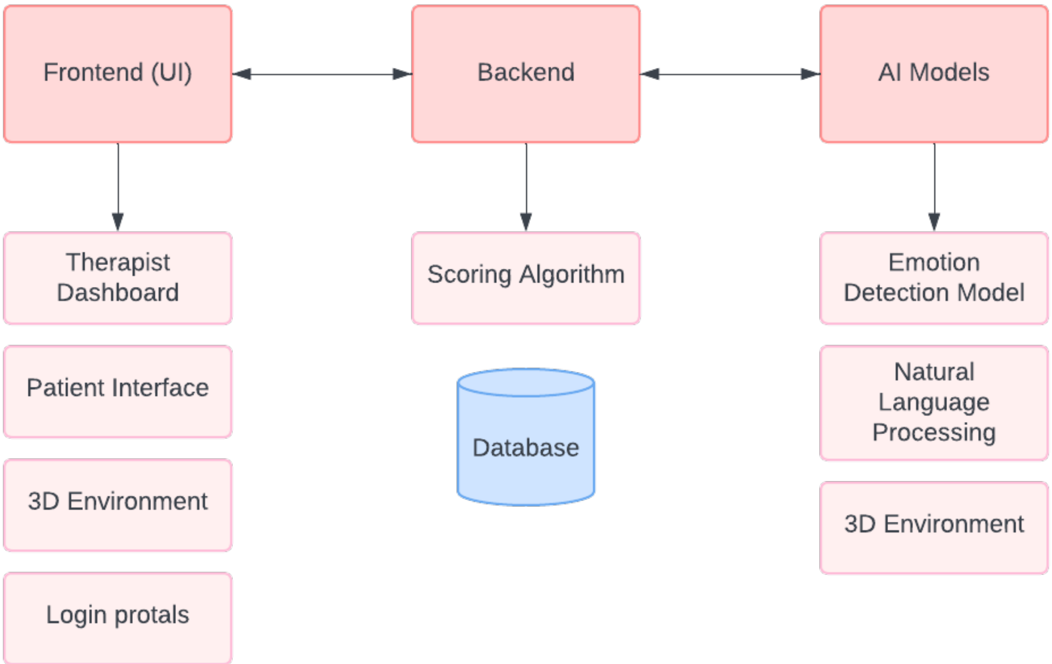

Figure 3.1: Box-and-Line High Level Architecture Diagram

- Emotion Detection (using PyTorch, analyzing facial expressions)
- Natural Language Processing (NLP) for understanding patient interactions

3.1.1 Relationships between modules

TherapEase consists of a multiple modules interacting with another, while fulfilling a specific function in implementing the whole system. Here's how these modules take part in creating one cohesive system:

1. **Frontend to Backend Interaction**
   The Frontend is the main User Interface through which both therapists and patients interact with the system. It communicates with the Backend to ensure data is fetched, updated, and stored properly.

   - **Login:** When a therapist or patient logs in, the Frontend Login sends the authentication details to the backend. The backend verifies the credentials and grants access, sending back session tokens for secure interaction throughout the system.
   - **Therapist Dashboard:** After login, therapists access their Dashboard, where they can view the patient progress, session history, and emotion analysis re-

sults. It requests data from the backend, which retrieves session logs and scores from the Database.

- **Patient Interactive Environment:** Patients interact with the 3D digital twin to answer the questions. These interactions are logged, and the data is processed and used for further AI analysis.

2. **Backend and AI Modules**
The AI Models analyzes the patient data, particularly focusing on emotional detection.

    - **Emotion Detection Model:** As the patient interacts with the 3D Environment, their facial expressions are captured through webcam. The Backend sends this data to the AI module, which uses PyTorch models to detect emotions like happiness, sadness, anger, surprise, fear, and disgust. The results are sent back to the Backend, where they are logged in the Database and used for scoring.
    - **ADDS Scoring:** The ADDS Scoring Model processes patient interaction data (e.g. responses to tasks) to calculate diagnostic scores.
    - **NLP Model:** If the patient engages in verbally, the NLP Model processes this data, which might also aid in emotional analysis.

3. **AI Models to Backend and Database** Once the AI Models have completed their analysis, the results are sent back to the Backend, where the Session Data is updated in the Database.

3.2 Design Models

Design Models for Object Oriented Development Approach

The applicable models for our project using object oriented development approach includes:

- Activity Diagram
- Deployment Diagram
- Sequence Diagram

3.2.1 Activity Diagram

The purpose of this activity diagram is to show the overall workflow of the system from the perspective of the patient and therapist.

- **Flow for the patient:**
    1. The patient logs into the system using the login page.
    2. The 3D digital twin is started for interaction with the patient.
    3. The digital twin asks questions, and the patient responds.
    4. The patients emotions during the interaction are monitored using webcam.
    5. All the emotional and response data are stored in the system's database.
    6. The patient logs out after the session.
- **Flow for the therapist:**
    1. The therapist logs into the system using the login page.
    2. The therapist has the option to either add, view or delete the patient data.
    3. In the view patient option, they can view all the session data and scores of a certain patient.
    4. After reviewing the session, the therapist logs out.

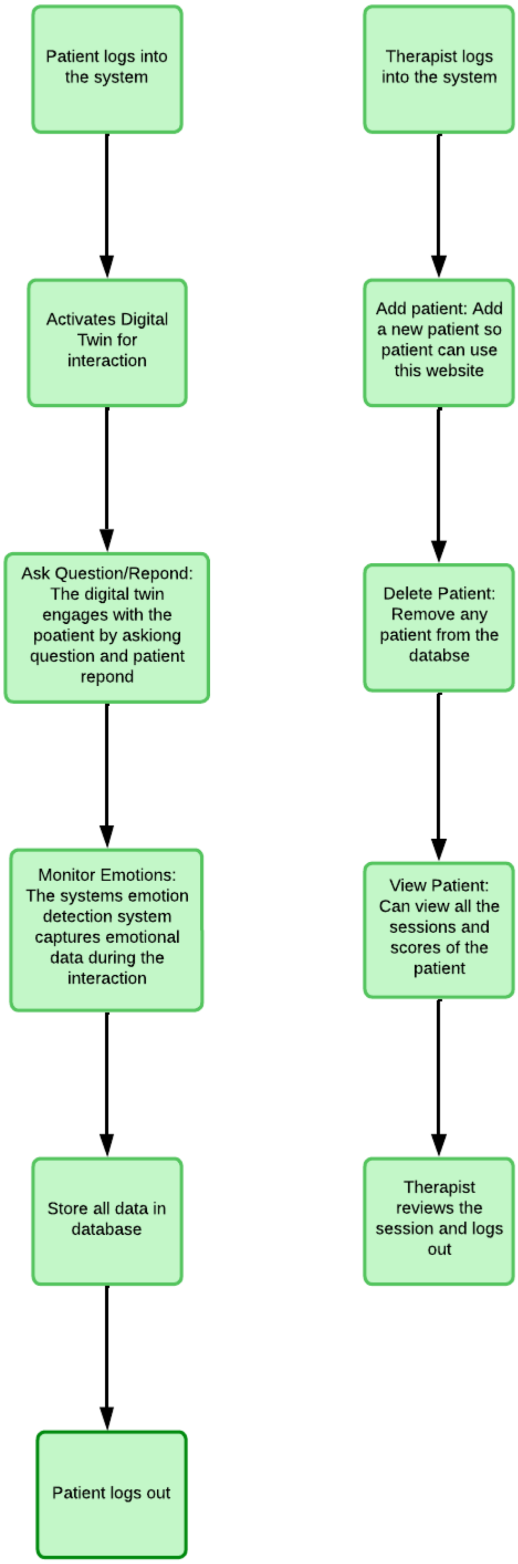

Figure 3.2: Activity Diagram

3.2.2 Deployment Diagram

The purpose of this deployment diagram is to explain the deployment of the software components and how the system is structured. The therapist uses a computer to manage patient profiles and views session reports. The patient uses another device to interact with the digital twin. All the patient data is stores and emotions are processed using an AI based emotion detection system.

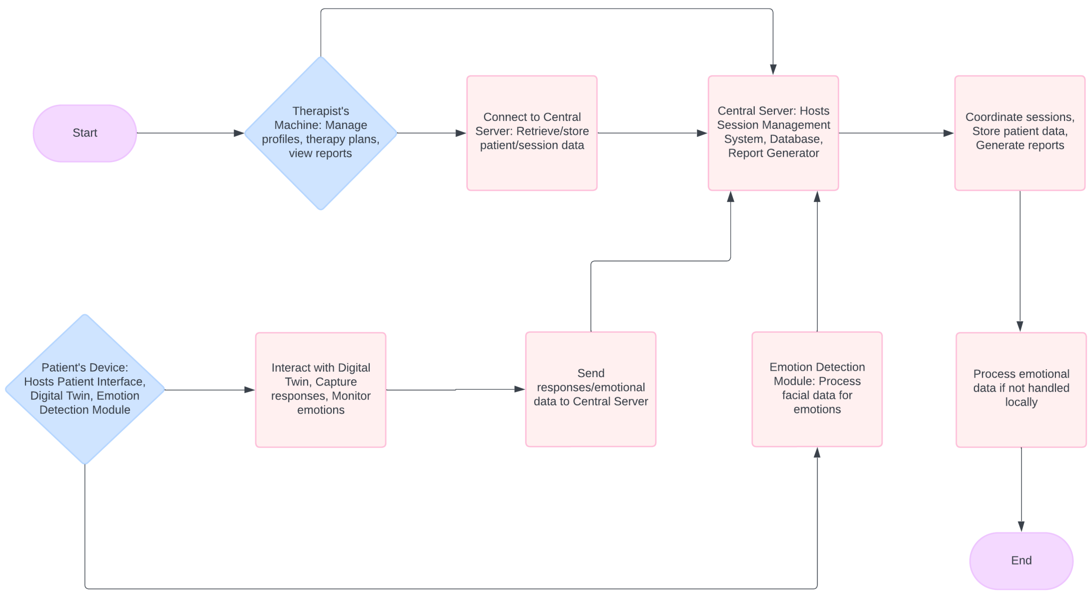

Figure 3.3: Deployment Diagram

3.2.3 Sequence Diagram

The purpose of this sequence diagram is to represent the step-by-step actions between the different components in the system during a session. It also shows how users (therapists and patients), the system, the digital twin, and devices interact in sequence.
The **therapist** logs in, activates the system for the **patient**. The patient interacts with a **digital twin** that asks questions, and the **system** tracks the patient's emotional responses through the **webcam**. All the data is recorded and stored, and the therapist reviews the results.

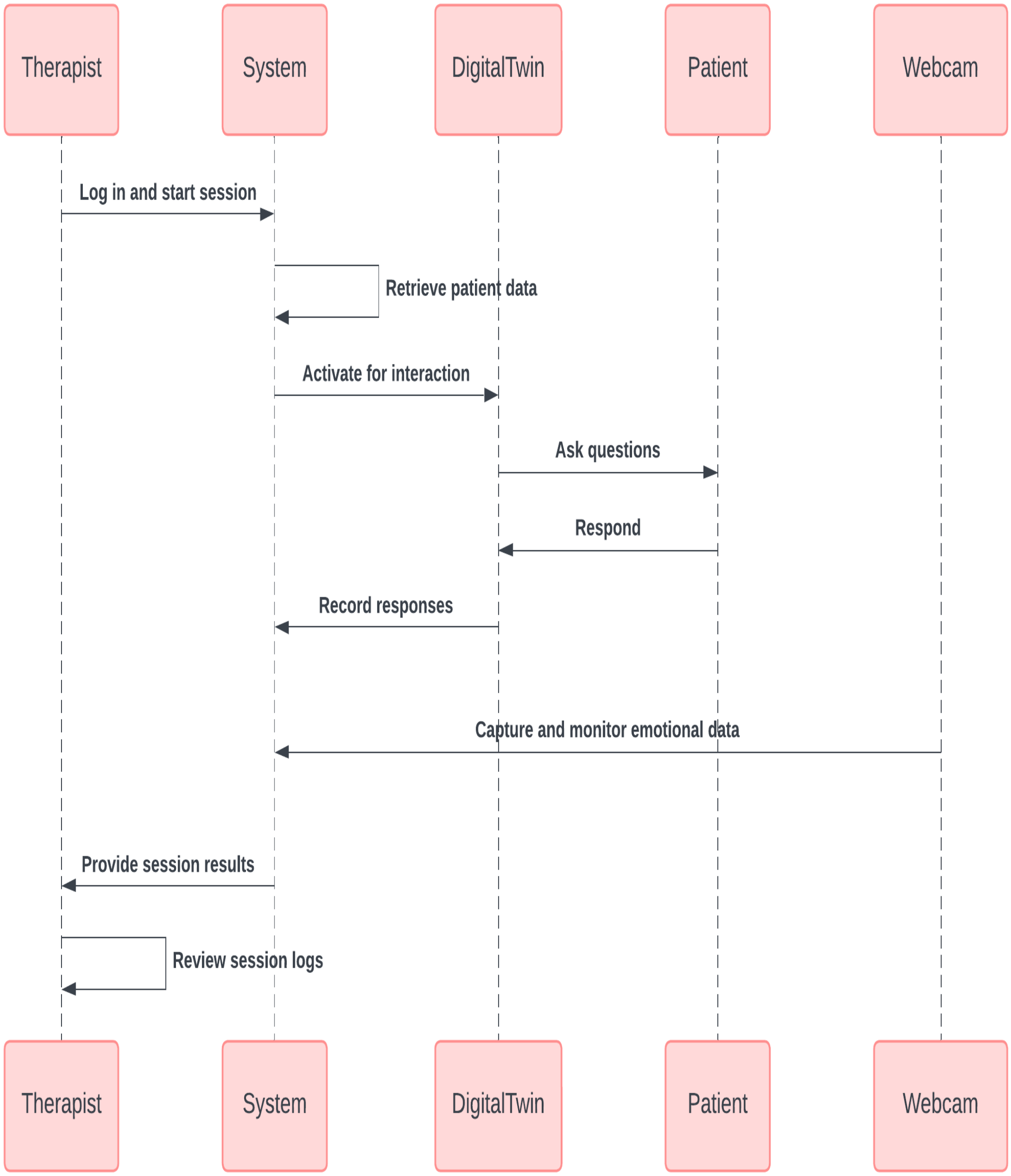

Figure 3.4: Class Level Sequence Diagram

3.3 Data Design

The design models for our project approach includes:

- Class Diagram
- Entity-Relationship diagram

3.3.1 Class Diagram

This class diagram represents a therapy system designed for autistic individuals, showcasing how key components like the therapist, patient, digital twin, and supporting systems work together. It enables personalized therapy, tracks patient progress, and allows for real-time adjustments to therapy plans based on emotional analysis. The database manager is essential for securely storing and accessing all patient and therapist information, improving the therapy experience and promoting ongoing enhancements in therapy delivery.

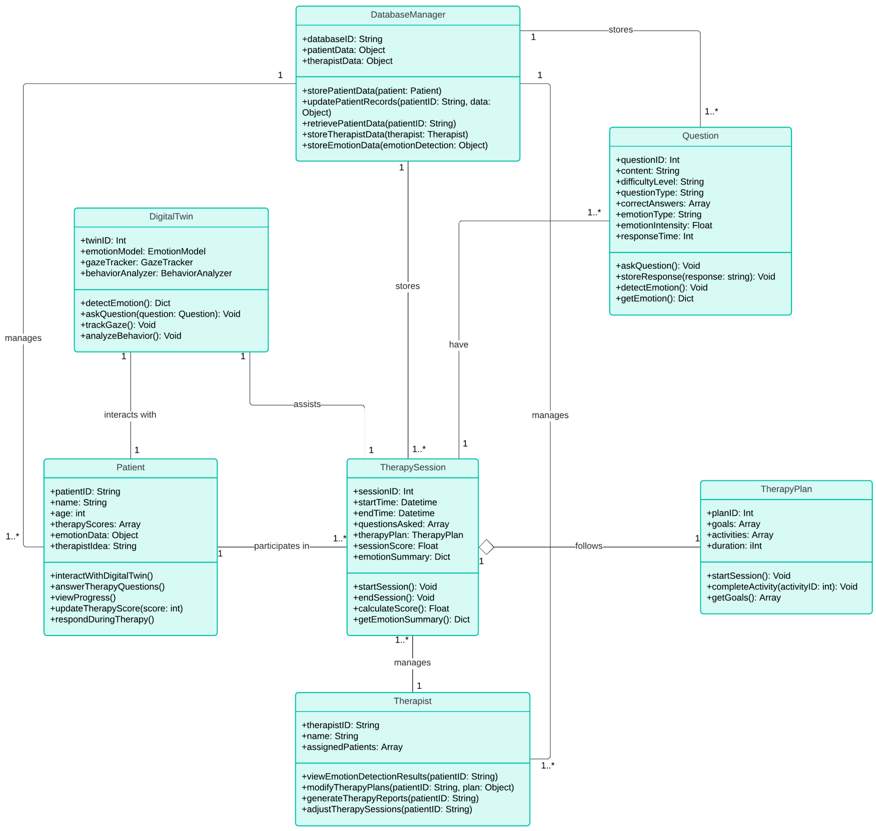

Figure 3.5: Class Diagram

3.3.2 Entity-Relationship Diagram

The purpose of this to represents how various tables are connected in a system so we can manage therapy sessions and track patient progress efficiently.

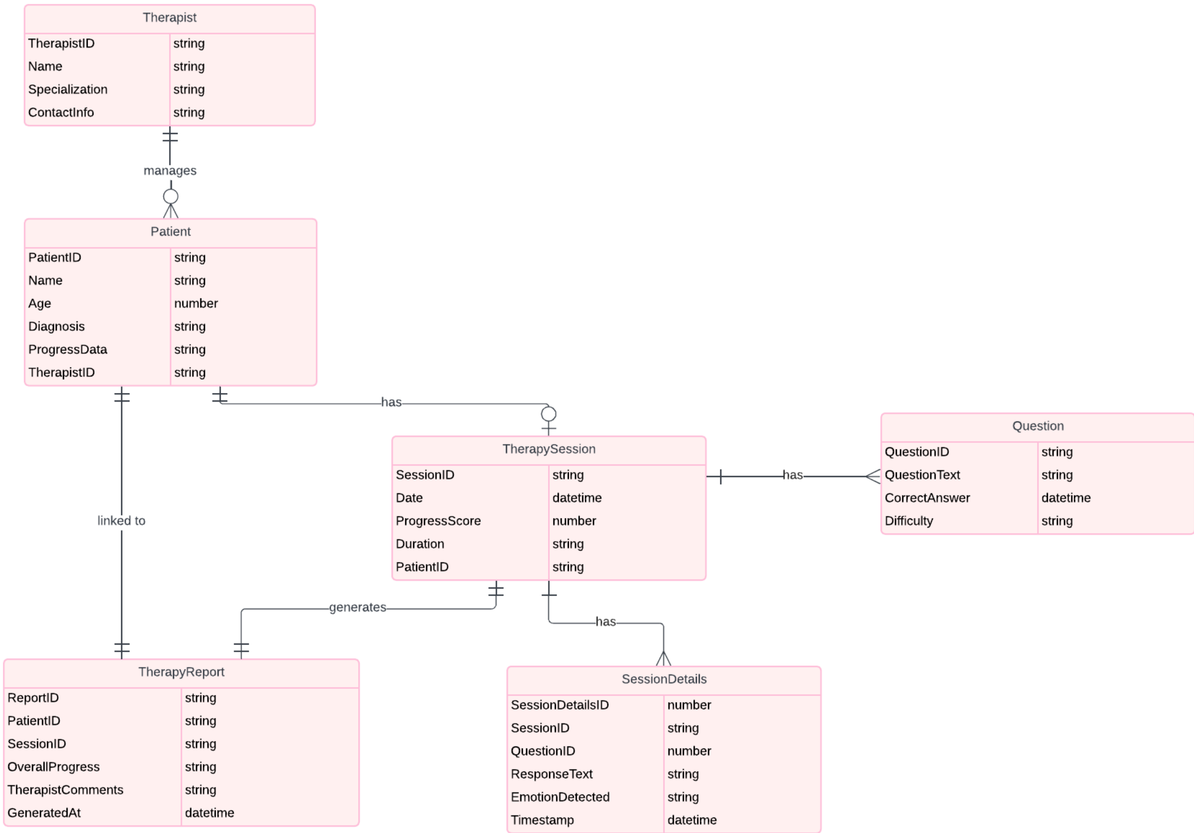

Figure 3.6: Entity-Relationship diagram

1. **Therapist Table**
   - Each therapist has a unique therapistID (primary key), along with other details like their name, years of experience, and qualifications.
2. **Patient Table**
   - This table stores information about patients individuals.
   - Each patient has a unique patientID (primary key), their name, age, and diagnosis details.
   - Each patient is assigned to a therapist, which is represented by the attribute assignedTherapist (foreign key).

3. **Session Table**

    - Each session has a unique sessionID (primary key), and is linked to a patient through the patientID (foreign key).
    - It records the date and duration of the session, as well as the patient's progress score after the session.

4. **Questions Table**

    - The questions table stores the therapy-related questions that are asked during a session along with the answers.
    - Each question has a unique questionID (primary key) and includes the questionText, the correctAnswer, and optionally, the difficulty level of the question.

5. **SessionDetails Table**

    - This table links to the session (sessionID) and questions (questionID) to record the patient's responses during a session along with emotions detected and the time stamp.

6. **TherapyReport Table**

    - Each report has a unique reportID (primary key) and is linked to the patient table (patientID), session table (sessionID), and therapist table (therapistID) using their foreign keys.
    - It also stores the overall progress, therapist comments, and the date when the report was generated.

Bibliography

[1] Accel Therapies. Aba assessment, n.d. [Online; accessed September 9, 2024].

[2] How to ABA. Aba assessment, n.d. [Online; accessed September 9, 2024].